Assignment 1-1 Practical Data Mining

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Task 1: Classification

1.1 Default Classifiers Comparison

Run No	Classifier	Parameters	Training Error	CV Error	Overfitting
1	ZeroR	Default	37.5%	37.5%	None
2	OneR	Default	7.5%	8.00%	None
3	J48	Default	0.5%	1.00%	None
4	IBk	Default	2.5%	4.25%	None

ZeroR is a trivial baseline that always predicts the majority class (**ckd,** 250/400 = 62.5% of instances). This establishes the floor: *any useful classifier must beat 37.5% error.* **OneR** learns a single-attribute rule. Here in our case, it selected hemoglobin (hemo) with the threshold < 13.25 → ckd.

J48 (C4.5 decision tree with pruning) built a tree of 9 leaves using 4 attributes (sc, pe, dm, hemo, sg). It misclassified only 2/400 training instances (0.5% error) and 4/400 under 10-fold CV (1.0% error). The gap is very small (0.5%), indicating that J48's pruning mechanism successfully controlled overfitting.

IBk (k-Nearest Neighbour with k=1 by default). With k=1, the training error is bounded by duplicate/near-duplicate instances — here 2.5%. However, under CV the error rises to 4.25%, giving the largest overfitting gap (1.75%) of the four classifiers. This is expected behavior for k=1.

1.2 J48 Parameter Tuning

Run No	Classifier	Parameters	Training Error	CV Error	Overfitting
1	J48	C=0.25; M=2	0.5%	1.0%	None
2	J48	C=0.10; M=2	0.5%	1.0%	None
3	J48	C=0.01; M=5	0.5%	1.5%	None
4	J48	C=0.001; M=10	3.0%	4.0%	None
5	J48	C=0.25; M=20	2.0%	3.0%	None

Based on the experimental runs, the optimal parameter combination for this dataset is C=0.25; M=2 (Run 1) or C=0.10; M=2 (Run 2). Both combinations yield the highest predictive accuracy with a Cross-validation Error of just 1.0% and minimal discrepancy between training and testing data.

1.3 Training Set Size Effect

Based on the below table, the training set grows from 10% to 100%, cross-validation error drops from 7.50% → 7.00% → 4.50% → 2.67% → 1.00%. This is the classic learning curve effect: *more examples give the model a more representative sample to learn from.*
In the 25% case, the largest overfitting gap (6.00%). With only 100 instances, J48 builds a complex 9-leaf tree that fits the training sample closely but fails to generalize. For this dataset, at least 50% of the data (200 instances) is needed for J48 to generalize reliably. Below this threshold, overfitting risk is high and the tree structure is unstable. For production use, the full dataset (400 instances) is preferable.

Training Size	Instances	Tree Leaves	Training Error	CV Error	Overfitting Gap
10%	40	2	7.50%	7.50%	0.00%
25%	100	9	1.00%	7.00%	**6.00%**
50%	200	4	4.00%	4.50%	0.50%
75%	300	8	1.67%	2.67%	1.00%
100%	400	9	0.50%	1.00%	0.50%

1.4 IBk Parameter Tuning

Note: In experiments, we only modified the K values of IBK and use default value for others

K	Training Error	CV Error	Overfitting Gap
1	2.50%	4.25%	1.75%
3	**4.25%**	**5.25%**	**1.00%**
5	5.00%	6.25%	1.25%
10	7.75%	9.25%	1.50%
15	11.50%	13.50%	2.00%
20	13.50%	14.50%	1.00%
30	17.25%	19.50%	2.25%
50	24.50%	26.50%	2.00%

Based on the experimental runs, the optimal parameter for the **IBk** classifier is **k=3**. This value successfully minimizes the overfitting gap to just **1.00%** while maintaining a highly accurate model with a Cross-validation Error of only **5.25%**.

1.5 Alternative Classifiers Evaluation

Run No	Classifier	Parameters	Training Error	CV Error	Overfitting Gap
1	Naive Bayes	Defaults: Gaussian numeric, no discretize	5.25%	5.00%	-0.25%

2	Naive Bayes	-K	2.25%	2.00%	-0.25%
3	Naive Bayes	-D	1.50%	2.00%	+0.50%
4	Random Forest	I=100; K=0; M=1; S=42	0.00%	0.25%	0.25%
5	Random Forest	I=200; K=0; M=1; S=42	0.00%	0.25%	0.25%

The best is **Random Forest (Runs 4 & 5):** Random Forest achieved near-perfect predictive accuracy with a Cross-validation (CV) Error of only **0.25%** and a Training Error of **0.00%**. Because it builds an ensemble of many decision trees (I=100 and I=200) and averages their predictions, it is highly robust. While the worst is **Default Naive Bayes (Run 1)**: With default settings, Naive Bayes assumes all numeric attributes follow a normal (Gaussian) distribution. In real-world medical data like chronic kidney disease, this is rarely true. Consequently, it struggled the most, yielding the highest CV Error of 5.00%.

1.6 ZeroR, OneR, and J48 Comparison

Based on the table result of 1.1, the three models, **J48** is vastly superior, achieving a near-perfect Cross-validation (CV) Error of **1.00%**. **OneR** serves as a surprisingly strong middle-ground model with an **8.00%** CV Error, while **ZeroR** performs the poorest with a **37.5%** CV Error.

- **ZeroR (The Baseline - 37.5% CV Error)**: ZeroR is practically blind; Because its error is 37.5%, this mathematically proves that 62.5% of your dataset belongs to the majority class. ZeroR acts as your absolute baseline.
- **OneR (The Single-Feature Rule - 8.00% CV Error)**: OneR looks at all available attributes but is restricted to building a rule based on only the single best predictor. The massive drop in error from 37.5% down to 8.00%.
- **J48 (The Multivariate Tree - 1.00% CV Error)**: J48 achieves the highest accuracy because it calculates Information Gain to build a complex decision tree using multiple attributes recursively. Instead of stopping at the single best feature like OneR, J48 splits the data, looks at the subsets. This allows it to capture complex, multi-variable. relationships.

1.7 Subgroup Testing

Subgroup definitions:

Case	**Subgroup**	**Split Rule**	**Confirmed**	**Missing attribute**	**Total**
Age	Young	age ≤ 30	51	+9 missing age	60

Age	Mature	$31 \leq age \leq 60$	208	+9 missing age	217
Age	Elder	age > 60	132	+9 missing age	141
BP	Normal BP	bp < 90 mmHg	304	+12 missing bp	316
BP	High BP	$bp \geq 90$ mmHg	84	+12 missing bp	96

In those experiments below we use classifier J48 (default parameters: -C 0.25 -M 2)

Case 1 — Age Groups:

Subgroup	Instances	Tree Leaves	Key Attributes Used	Training Error	CV Error	Overfitting Gap
Full dataset (ref)	400	9	sc, pe, dm, hemo, sg	0.50%	1.00%	0.50%
Young (age ≤ 30)	60	7	hemo, pe, sg	1.67%	1.67%	0.00%
Mature (31–60)	217	8	hemo, sc, al	1.84%	5.07%	3.23%
Elder (age > 60)	141	4	sc, htn, rbc	2.84%	3.55%	0.71%

Case 2 — Blood Pressure Groups

Subgroup	Instances	Tree Leaves	Key Attributes Used	Training Error	CV Error	Overfitting Gap
Full dataset (ref)	400	9	sc, pe, dm, hemo, sg	0.50%	1.00%	0.50%
Normal BP (bp < 90)	316	9	sc, dm, hemo, pe, sg	0.32%	2.22%	1.90%
High BP (bp ≥ 90)	96	1	(none — majority class)	2.08%	2.08%	0.00%

Model performance varies significantly across subgroups, even though J48 achieves 99% CV accuracy on the full dataset. CV errors range from 1.67% (young) to 5.07% (mature) across age groups, and from 2.08% (high BP) to 2.22% (normal BP) across BP groups.

The mature age group (31–60) is the hardest to classify (5.07% CV error, 3.23% gap). Middle-aged patients present the most diverse clinical picture, with more ambiguous boundary cases.

High blood pressure is nearly a definitive indicator of ckd in this dataset. The high-BP subgroup has ~98% ckd prevalence, causing **J48** to degenerate into a trivial majority-class predictor. This has important clinical implications — patients presenting with high diastolic BP should be considered at a very high risk for ckd.

1.8 Attribute Selection

Attribute selection methods:

- **Method 1**: **CfsSubsetEval** + **BestFirst** - automatically selects the best correlated non-redundant subset
    - **Attributes kept (17):** bp, sg, al, rbc, bgr, bu, sc, sod, pot, hemo, pcv, wbcc, htn, dm, appet, pe, ane.
    - **Attributes removed (7):** age, su, pc, pcc, ba, rbcc, cad.
- **Method 2: InfoGainAttributeEval + Ranker (N=5) -** ranks all attributes by information gain, keeps the 5 highest
    - **Attributes kept (5)**: hemo, sc, sg, pcv, al
    - **Attributes removed (19)**: all others

J48 Results Comparison:

Dataset	Number of Attributes	Tree leaves	Tree Structure	Training Error	CV Error	Overfitting Gap
Full dataset	24	9	sc, pe, dm, hemo, sg	0.50%	1.00%	0.50%
Method 1	17	9	sc, pe, dm, hemo, sg	0.50%	1.00%	0.50%
Method 2	5	7	sc, hemo, sg	2.00%	2.00%	0.00

Despite the reduction in 7 of the 24 attributes in method 1, J48 produces an identical tree to the full dataset (the same leaves, structure, and error). Therefore, the selector removes 7 redundant attributes with zero accuracy loss. On the other hand, just 5 attributes in method 2 are sufficient for 98% CV accuracy. That reflect the attributes (hemo, sc, sg, pcv, al) which can capture the essential clinical signal.

For real application, the selector of method 1 (17 attributes) is the best choice while reducing data collection cost. If clinical simplicity is a priority, method 2 provides an excellent trade-off: near-perfect accuracy with only 5 attributes.

Task 2: Numeric Prediction

2.1 Default Models Comparison

Preprocessing data: Combining winequality-red.csv and winequality-white.csv to a single file for model training.

Table results:

Model	Parameter	Train Corr	CV Corr	Train MAE	CV MAE	Train RMSE	CV RMSE
ZeroR	Default	0.0000	−0.0365	0.6856	0.6857	0.8732	0.8733
M5P	Default	0.6322	0.5921	0.5299	0.5458	0.6767	0.7039
IBk	Default	1.0000	0.6086	0.0000	0.425	0.0000	0.7652

Comparison with ZeroR:

Model	Train RAE	CV RAE	Train RRSE	CV RRSE
M5P	77.30 %	79.60 %	77.50 %	80.60 %
IBk	0.00%	62.04%	0.00%	87.62%

ZeroR works as expected for the baseline model, which simply guesses the *average* quality score of the entire wine dataset for every single instance. **M5P** shows a solid predictive capability with a CV Correlation of 0.5921. More importantly, it does not overfit. The gap between its Training MAE (0.5299) and CV MAE (0.5458) is incredibly small, meaning the model generalizes very well to unseen data. But **IBk,** because Weka's default **IBk** uses K=1. it exhibits textbook overfitting. The IBK results show a perfect Training Correlation of 1.0000 and a Training MAE/RMSE of exactly 0.0000 because the model literally memorized the training set by looking only at the single nearest data point. Consequently, its performance degrades noticeably on the CV test, proving it memorizing the noise rather than learning the actual patterns.

Out of the default models, **M5P** is the most robust and reliable predictor, while default **IBk** suffers from severe overfitting

2.2 M5P and IBk Parameter Tuning

After many experiments in appendix 1, achieve top 5 best runs (based on the CV RMSE)

Rank	Model	Params	CV Corr	CV MAE	CV RMSE	CV RRSE	RMSE Gap

1	IBK	K = 10, inverse-distance	0.7141	0.4163	0.6131	70.20%	+0.4772
2	IBK	K = 5, inverse-distance	0.6981	0.4114	0.6299	72.13%	+0.5468
3	IBK	K = 10	0.6083	0.5329	0.6958	79.67%	+0.0679
4	IBK	K = 5	0.6063	0.5240	0.7027	80.46%	+0.1341
5	M5P	-M 20 (min 20 per leaf)	0.5938	0.5424	0.7032	80.52%	+0.0597

While Rank 1 (**IBk**, K=10, inverse-distance) yields the highest CV Correlation, rank 3 (**IBk**, K=10, no distance weighting) is arguably the most optimal model because it strikes the best balance between predictive accuracy and minimizing the overfitting gap.

In rank 3 and rank 1, when turned off inverse-distance weighting and just usedandard majority vote among the 10 nearest neighbors, CV Correlation dropped to 0.6083. However, RMSE Gap plummeted to a tiny +0.0679. This means the model is incredibly stable and generalizes exceptionally well to unseen data, making it a safer, more reliable real-world predictor than rank 1.

2.3 Alternative Classifiers Evaluation

After many experiments in appendix 2, achieve top 5 best runs (based on the CV RMSE)

Rank	**Run**	**Parameters**	**CV Corr**	**CV MAE**	**CV RMSE**	**CV RRSE**	**RMSE Gap**
1	RF	200 trees, K (default)	0.7458	0.4129	0.5858	67.08%	+0.3721
2	RF	100 trees, K (default)	0.7438	0.4143	0.5873	67.25%	+0.3716
3	RF	100 trees, K=5	0.7427	0.4149	0.5880	67.32%	+0.3712
4	RF	50 trees, K=√11	0.7399	0.4175	0.5904	67.60%	+0.3694

5	RF	100 trees, K=11	0.7328	0.4208	0.5958	68.23%	+0.3770

In appendix 2 (full picture), a low overfitting gap does not mean a good model**. LR's** near-zero gap (train ≈ CV) means it is fitting the training data and the unseen data equally badly. The gap is small not because LR generalizes well, but because it has already hit a performance ceiling.

While RF's large gap is a sign of model richness, not a failure**.** RF's training **RMSE** is excellent but not achievable on unseen data because individual trees overfit their bootstrap samples. However, the CV RMSE is still far better than LR's CV RMSE.

Take the model rank 1 to perform with different wine (red and white)

Dataset	**Train Corr**	**CV Corr**	**Train MAE**	**CV MAE**	**Train RMSE**	**CV RMSE**	**Train RRSE**	**CV RRSE**
Full	0.9799	0.7458	0.1490	0.4129	0.2137	0.5858	24.48%	67.08%
Red wine	0.9780	0.7173	0.1469	0.4030	0.2079	0.5645	25.76%	69.85%
White wine	0.9796	0.7492	0.1493	0.4155	0.2168	0.5900	24.48%	66.60%

Red wine (1,599 instances) — CV RMSE: 0.5645 | CV Corr: 0.7173 | RRSE: 69.85%

As a result, it is harder for the model to distinguish between quality levels, and the relative performance gain is smaller. The CV correlation (0.7173) is the lowest of the three datasets, consistent with red wine quality being harder to predict relatively — the physicochemical features explain a smaller share of the quality variance in red wines than in white wines.

White wine (4,898 instances) — CV RMSE: 0.5900 | CV Corr: 0.7492 | RRSE: 66.60%

RF performs relatively best on white wine: RRSE = 66.60% (lowest), CV correlation = 0.7492 (highest), even though the absolute CV RMSE (0.5900) is higher than red wine.

Conclusion: **RF** generalizes better on white wine than red wine in relative terms (RRSE 66.60% vs. 69.85%), despite white wine's higher absolute RMSE. The larger white wine training set and stronger physicochemical-quality signal in white wine both contribute to better relative performance. Red wine quality prediction is fundamentally harder because its quality range is narrower and more subjectively influenced by non-measurable factors.

2.4 Feature Selection & Engineering

Feature Selection Impact

Method	Evaluator	Attributes kept
CFS + BestFirst	CfsSubsetEval	4: volatile acidity, citric acid, chlorides, alcohol
ReliefF top-5	ReliefFAttributeEval -K 10	5: alcohol, fixed acidity, volatile acidity, pH, sulphates
Correlation top-8	CorrelationAttributeEval	8: alcohol, citric acid, free SO_2, sulphates, pH, residual sugar, total SO_2, fixed acidity

Result tables

Model	Parameter	Dataset / method	CV Corr	CV MAE	CV RMSE	CV RRSE	vs Baseline RRSE
RF (baseline)	-I 200	Full 11 attributes	0.7458	0.4129	0.5858	67.08%	
LR (baseline)	Default	Full 11 attributes	0.5367	0.5698	0.7368	84.36%	
RF	-I 200	CFS + BestFirst	0.6610	0.4607	0.6556	75.07%	−7.99%
LR	Default	CFS + BestFirst	0.5087	0.5873	0.7518	86.08%	−1.72%
RF	-I 200	ReliefF top-5	0.6949	0.4418	0.6283	71.94%	−4.86 %
LR	Default	ReliefF top-5	0.5179	0.5819	0.7469	85.53%	−1.17%

RF	-I 200	Correlation top-8	0.7256	0.4282	0.6032	69.07%	−1.99 %
LR	Default	Correlation top-8	0.4851	0.5987	0.7636	87.44%	−3.08%

Note: the `vs Baseline RRSE` column is computed by ***CV RRSE (baseline)*** **– CV RRSE.** *If it's positive means improvement over full dataset otherwise negative = degradation*
For this wine quality dataset, attribute selection consistently degrades performance for both RF and LR. The 11 physicochemical attributes collectively contain more predictive signal than any strict subset.

Feature engineering Impact

Technique	Description
Normalization	Scale all 11 attributes to [0, 1]
Wine-type indicator	Add wine_type {red, white} as a new nominal attribute
PCA	PrincipalComponents -R 0.95 — retain components explaining ≥ 95% variance (supervised, uses class)

Result tables

Model	**Parameter**	**Dataset / method**	**CV Corr**	**CV MAE**	**CV RMSE**	**CV RRSE**	**vs Baseline RRSE**
RF (baseline)	-I 200	Full 11 attributes	0.7458	0.4129	0.5858	67.08 %	
LR (baseline)	Default	Full 11 attributes	0.5367	0.5698	0.7368	84.36 %	

RF	-I 200	Normalization	0.7444	0.4146	0.5871	67.22%	−0.14%
LR	Default	Normalization	0.5367	0.5698	0.7368	84.36%	0.00%
RF	-I 200	Wine-type indicator	0.7444	0.4142	0.5874	67.25%	−0.17%
LR	Default	Wine-type indicator	0.5404	0.5699	0.7347	84.13%	+0.23%
RF	-I 200	PCA (9 components)	0.7292	0.4254	0.6017	68.89%	−1.81%
LR	Default	PCA (9 components)	0.5370	0.5700	0.7366	84.35%	−0.01%

In **normalization**, normalizing the 11 input attributes to the [0, 1] range (the quality target is kept on its original 3–9 scale) has zero measurable effect on both **RF** and LR.

In **Wine-type indicator, a**dding an explicit wine_type {red, white} attribute (combining typed red + white ARFF files) had nearly no effect:

- **RF**: RRSE = 67.25% (−0.17% vs baseline — marginal degradation)
- **LR**: RRSE = 84.13% (+0.23% improvement — marginal gain)

Unsupervised PCA is neutral-to-slightly-harmful for these two models. It offers no accuracy gain over using all 11 raw attributes, confirming that the original feature space is already well-conditioned for tree splits.

Overall Conclusions

- **Feature selection uniformly hurts performance** for this wine dataset. All three attribute-selection methods (CFS, ReliefF, Correlation) reduce CV RRSE for RF, with the number of removed attributes determining the degree of degradation.
- **Feature engineering** (normalisation, wine-type) provides negligible benefit for Random Forest. RF is scale-invariant and already learns wine-type boundaries from the feature distributions. These transformations add no new information the model cannot already derive.
- **The best practical recommendation** is to retain all 11 physicochemical attributes and use RF (-I 200) with the wine-type as a stratification variable (separate models for red and white), rather than a mixed dataset with a type of indicator.

Appendix

Appendix 1 – Full experiments in task 2.2

Run	Model	Parameter	Train Corr	CV Corr	Train MAE	CV MAE	Train RMSE	CV RMSE
1	M5P	-M 4 (default, smoothed)	0.6322	0.5921	0.5299	0.5458	0.6767	0.7039
2	M5P	-M 10	0.6727	0.5914	0.5015	0.5429	0.6468	0.7048
3	M5P	-M 20	0.6769	0.5938	0.4925	0.5424	0.6435	0.7032
4	M5P	-M 50	0.6547	0.5833	0.5067	0.5476	0.6604	0.7097
5	M5P	-U (no smoothing)	0.6334	0.5870	0.5289	0.5444	0.6757	0.7083
6	M5P	-R (regression tree)	0.6221	0.5719	0.5258	0.5515	0.6845	0.7163
7	M5P	-M 10 -U	0.6800	0.5823	0.4912	0.5447	0.6403	0.7128
8	IBk	K = 1 (default)	1.0000	0.6086	0.0000	0.4254	0.0000	0.7652

9	IBk	K = 3	0.8224	0.6023	0.3295	0.5094	0.4971	0.7153
10	IBk	K = 5	0.7595	0.6063	0.4141	0.5240	0.5686	0.7027
11	IBk	K = 10	0.6964	0.6083	0.4797	0.5329	0.6279	0.6958
12	IBk	K = 5, -I (inv dist)	0.9968	0.6981	0.0569	0.4114	0.0831	0.6299
13	IBk	K = 10, -I (inv dist)	0.9921	0.7141	0.0986	0.4163	0.1359	0.6131
14	IBk	K = 5, -X (auto-K)	1.0000	0.6086	0.0000	0.4254	0.0000	0.7652

Appendix 2 – Full experiments in task 2.3

Run	Model	Parameters	Train Corr	CV Corr	Train MAE	CV MAE	Train RMSE	CV RMSE
1	LR	-S 0 (M5 selection, ridge=1e-8)	0.5403	0.5367	0.5685	0.5698	0.7348	0.7368
2	LR	-S 1 (all 11 attrs, no selection)	0.5405	0.5372	0.5683	0.5694	0.7347	0.7365
3	LR	-S 2 (greedy hill-climbing)	0.5403	0.5367	0.5685	0.5698	0.7348	0.7368
4	LR	-R 1.0 (strong ridge)	0.5403	0.5367	0.5685	0.5698	0.7348	0.7368
5	LR	-R 0.001 (mild ridge)	0.5403	0.5367	0.5685	0.5698	0.7348	0.7368

6	RF	100 trees, K=√11 features	0.9789	0.7438	0.1490	0.4143	0.2157	0.5873
7	RF	-I 50 (50 trees)	0.9767	0.7399	0.1516	0.4175	0.2210	0.5904
8	RF	-I 200 (200 trees)	0.9799	0.7458	0.1490	0.4129	0.2137	0.5858
9	RF	-K 5 (fixed 5 features/split)	0.9786	0.7427	0.1505	0.4149	0.2168	0.5880
10	RF	-K 11 (all features/split)	0.9775	0.7328	0.1521	0.4208	0.2188	0.5958

Appendix 3 – Full experiments and reports

Experiments and Reports